

# CAPSTONE PROJECT SUBMISSION

## ACADEMIC INTEGRITY DECLARATION

*AI/ML Fundamentals Course*

This declaration must be completed and signed by the student. The project submission will not be considered complete without this confirmation.

### Student Declaration

By signing this cover sheet, I confirm that:

- The submitted Capstone Project is my own work.
- I have not copied another student's project, code, report, documentation, or results.
- All external datasets, libraries, pretrained models, code samples, AI tools, and other sources used in the project are clearly acknowledged in the repository.
- I have not fabricated, altered, or misrepresented evaluation metrics, experiment results, screenshots, logs, or repository history.
- I have not presented an external AI service or hosted API as a model trained or fine-tuned by me.
- I understand the submitted code and can explain and defend the project, its pipeline, and its results.
- The repository link included in the submission form points to the final version of the project, contains the reproducible demo/Colab workflow, and is accessible to the mentors.

I understand that plagiarism, fabricated results, unauthorized collaboration, or misrepresentation of authorship may result in a score of 0 for the Capstone Project and may lead to additional academic disciplinary action.

I have read, understood, and agree to the declaration above.

|                     |                                       |
|---------------------|---------------------------------------|
| Full Name *         | Mubinabegim To'lqinjonova             |
| Date *              | 01-08-2026                            |
| Student Signature * | I confirm - Mubinabegim To'lqinjonova |

*Complete this cover sheet before filling in the project submission form on the next page.*

# CAPSTONE PROJECT SUBMISSION FORM

AI/ML Fundamentals Course

Instructions: Download this file, complete all required fields, save it as PDF or DOCX, and upload it to the LMS. The repository must be accessible to the mentors before the deadline.

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Full Name *                      | Mubinabegim To'lqinjonova                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2. Project Track *                  | <input type="checkbox"/> Individual Project Track<br><input checked="" type="checkbox"/> <b>Field-Based Scenario Track</b><br>Scenario EDU-02: Student Performance Early-Warning Model (EdTech)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3. Project Title *                  | Student Performance Early-Warning Model: predicting academic risk from early-course data (OULAD)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4. Repository URL *                 | <a href="https://github.com/Mubina-lazy/edu02-early-warning-model">https://github.com/Mubina-lazy/edu02-early-warning-model</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5. Short Project Description *      | <p><b>Problem addressed:</b><br/>Universities usually identify struggling students only after final results are recorded, when it is too late to intervene. This project estimates, after roughly the first quarter of an online course, whether a student is at risk of failing or withdrawing, so that support can still change the outcome.</p> <p><b>Expected user or stakeholder:</b><br/>Academic advisors and student-support staff. They receive a ranked list of students to contact while there is still time to help. The output is decision support only and every flag requires human review; it must not drive admission, grading or disciplinary decisions.</p> <p><b>ML task and main output:</b><br/>Supervised binary classification on the OULAD dataset (32,593 student enrolments across 22 course presentations, CC-BY 4.0). The model sees only information available at 25% of the course presentation: demographics, virtual learning environment activity, and scores of assignments already due. For each student it outputs a risk probability, a Low/Medium/High band, a flag decision at a threshold tuned on validation (0.327), and the top contributing factors taken from the model itself (TreeSHAP). Final model: XGBoost, selected from seven MLflow-tracked runs. On the held-out 2014J cohort it reaches recall 0.760, precision 0.583, F1 0.660 and PR-AUC 0.714 for the at-risk class, against 0.000 recall for a majority-class baseline.</p> |
| 6. Optional Demo / Application Link | Reproducible Google Colab demo (opens demo.ipynb from the repository and runs as-is; no dataset download needed):<br><a href="https://colab.research.google.com/github/Mubina-lazy/edu02-early-warning-model/blob/main/demo.ipynb">https://colab.research.google.com/github/Mubina-lazy/edu02-early-warning-model/blob/main/demo.ipynb</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7. Access Instructions              | Not required. The repository is public, so no additional steps, credentials or artifacts are needed. The trained model ships with the repository, so the Colab demo runs without downloading the dataset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8. Repository Access Confirmation * | <input checked="" type="checkbox"/> I confirm that the repository is accessible to the mentors, contains the final project version, and includes the reproducible demo/Colab workflow or clear instructions to run it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |